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ADITYA ENGINEERING COLLEGE

ADB ROAD, SURAMPALEM, E.G. DIST.

DEPARTMENT OF

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Certified that this is the bonafide record of practical work done by

Mr. / Ms. .....

a students of ..... with PIN No.....

in the ..... Laboratory during the year .....

No of Practicals Conducted [ ] No. of Practicals Attended [ ]

Signature - Faculty Incharge Signature - Head of the Department

Submitted for the Practical examination held on.....

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2.		Using UML/JAVA, design Abstract Factory design pattern.	
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5.		Using UML/JAVA, design Bridge design pattern.	
6.		Using UML/JAVA, design Decorator design pattern.	
7.		User gives a print command from a word document. Design to represent this chain of responsibility design pattern.	

Experiment – 1 : Take any real time problem and do the following experiments.

Finding part-time jobs(Find Your Opportunity) involves thorough research to identify suitable opportunities through online job boards, company websites, and local listings. Once identified, it's crucial to update and tailor your resume to highlight relevant skills and experiences. Crafting a concise cover letter that explains your interest in part-time work and aligns your qualifications with the job requirements is essential. Submitting applications promptly and following up afterward demonstrates initiative and interest. Networking within your industry or community can also uncover hidden job opportunities. Finally, preparing for interviews by practicing common questions and dressing professionally enhances your chances of securing a part-time position. By diligently following these steps, you can effectively navigate the process of finding and applying for part-time jobs.

Date: _ _ _ _ _ _ _ _ _ _ _

Experiment – 1.a: Write down the problem statement for a suggested system of relevance. Develop Flow-Charts to understand basic problem solving technique.

Problem Statement

In today's dynamic job market, individuals seeking part-time employment face challenges in efficiently identifying suitable job opportunities that match their skills, preferences, and geographic location. Traditional job search methods often rely on manual browsing of multiple job portals, leading to inefficiencies and missed opportunities. Moreover, the sheer volume of job listings available online necessitates a more systematic approach to streamline the search process.

why I want to work on this problem:

Working on this problem offers an opportunity to combine personal experience, professional insight, and broader significance to drive motivation and passion for finding innovative solutions in the realm of task management.

Personal Experience:

I successfully found a part-time job by diligently researching opportunities, tailoring my applications, and leveraging networking to secure interviews and eventually land a position in customer service.

Professional Insight:

Professional insights into finding part-time jobs emphasize strategic research, tailored applications, and effective networking to align skills with job requirements and showcase reliability and enthusiasm, enhancing chances of securing positions through persistence and professionalism.

Broader Significance:

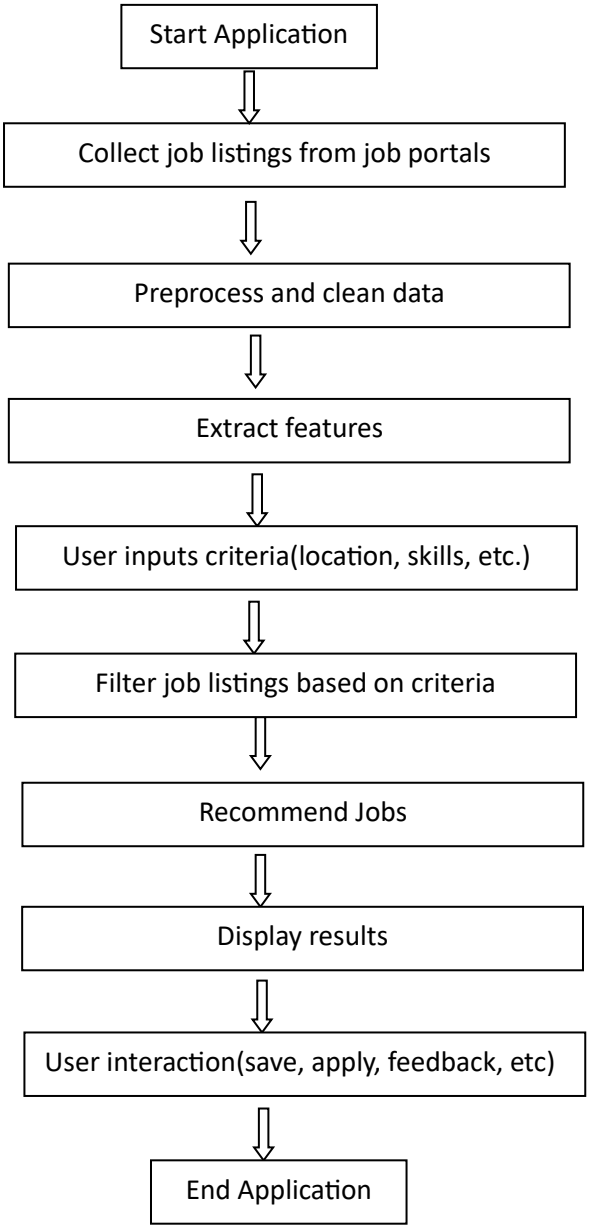
Borders hold significant economic, cultural, and political importance, serving as conduits for trade and commerce, markers of cultural identity, and boundaries that define sovereignty and international relations. They symbolize both connection and division, influencing geopolitical landscapes and shaping global interactions.

Motivation:

Passion for Productivity and Goal Achievement

Desire to Make a Meaningful Difference

FLOW-CHART


Start Application
Collect job listings from job portals
Preprocess and clean data
Extract features
User inputs criteria(location, skills, etc.)
Filter job listings based on criteria
Recommend Jobs
Display results
User interaction(save, apply, feedback, etc)
End Application

Experiment – 1.b: Do requirement analysis and develop Software Requirement Specification Sheet (SRS) for suggested system.

Introduction:

In today's dynamic job market, individuals seeking part-time employment often face challenges in finding suitable opportunities that match their skills schedule, and location preferences. Traditional method of job searching may be time consuming and in-efficient, leading to frustration and missed opportunities. Additionally, the lack of centralized platform catering specifically to part-time job seekers exacerbates this issue, making it difficult for individual to discover and apply for relevant positions.

Requirements:

Functional Requirements:

1.Data Collection and Integration:

- The system shall collect part-time job listings from at least three major job portals (e.g., Indeed, LinkedIn, Glassdoor) using their APIs.
- The system shall support automated data retrieval at regular intervals to ensure job listings are up-to-date.

2. Data Processing and Analysis:

- The system shall clean and preprocess raw job data to remove duplicates, standardize formats, and enhance data quality.
- The system shall extract relevant features from job listings including job title, location, salary, required skills, and job description keywords.

3. Job Recommendation and Filtering:

- The system shall allow users to input criteria such as location, industry, pay range, and required skills to filter job listings.
- The system shall implement keyword matching algorithms to match user-defined criteria with job listings.

4. User Interface (UI/UX):

- The system shall feature a user-friendly web or mobile interface allowing job seekers to easily navigate and interact with job listings and visualizations.
- The system shall provide personalized job recommendations and allow users to save favorite job listings or apply directly through integrated links.

5. Security and Privacy:

- The system shall implement secure data transmission and storage practices to protect user information and job listing data.
- The system shall comply with relevant data privacy regulations (e.g., GDPR, CCPA) regarding user consent and data handling.

Non-Functional Requirements:

1. Performance:

- The system shall respond to user queries and interactions within a maximum of 3 seconds, under normal operating conditions.
- The system shall handle concurrent requests from at least 100 users without significant degradation in performance.

2. **Reliability:**

- The system shall have a system uptime of at least 99.9% excluding scheduled maintenance windows.
- Data backups shall be performed daily and stored securely to prevent data loss in case of system failure or corruption.

3. **Scalability:**

- The system architecture shall be designed to scale horizontally to handle increased data volume and user traffic as the user base grows.
- The system shall support adding new job portals or data sources without significant downtime or disruption to existing functionality.

4. **Security:**

- User authentication and authorization shall be implemented using industry-standard protocols (e.g., OAuth 2.0) to ensure secure access to user data.
- All data transmissions between the client and server shall be encrypted using HTTPS to prevent unauthorized access and data breaches.

5. **Usability:**

- The user interface shall be intuitive and accessible, ensuring ease of navigation and interaction for users of varying technical expertise.
- The system shall provide clear error messages and feedback to users in case of invalid inputs or system errors.

6. **Compatibility:**

- The system shall be compatible with major web browsers (e.g., Chrome, Firefox, Safari) and mobile platforms (iOS, Android).
- The system shall support integration with popular third-party tools and APIs used for analytics and reporting.

Experiment – 1.c: Using COCOMO model estimate effort.

Basic COCOMO Model:

Assumptions

Project Size :10000 lines of code (LOC)

COCOMO Model: Organic

Error Adjustment Factor (EAF) :1.0(Neutral Adjustment)

Effort = a x (size) pow(b) x EAF

a=2.4

b=1.05

size = 10000 (Loc)

EAF = 1.0

Effort = 2.4 x (10000) pow (1.05) x 1.0

Effort = 2.4 x (10) pow (1.05)

Effort = 2.4 x 11.047

Effort = 26.5128 persons-months

Time (T) = C (E) pow (d)

= 2.5 (26.5128) pow (0.38)

= 2.5 x 7.6477

T = 19.11925

INTERMEDIATE MODEL:

Project Size: 20000 lines of code(loc)

COCOMO model: Organic

Effort Adjustment Factor (EAF): Detailed Factors Considered

Effort = a x (size) pow (b) x EAF

a = 2.4

b = 1.05

EAF = 1.2

Size = 20000

Where EAF is effort is calculated as;

EAF = (Summation I = 1 to N Rating I)

Where 'n' is the total numbers of cost divers and the rating 'I' is the rating assigned to each cost divers and size is the estimated project size in KLOC.

From, Product Attribute (Nominal)

The Complexity of the product = 1.00

Hardware Attribute (High)

The Required Turn-Around Time = 1.07

Project Attributes (Low)

Use of Software Tools = 1.10

EAF = 1.00 x 1.07 x 1.10

EAF = 1.177

EFFORT (E) = 1.177 x 26.5128

E = 31.2055 Persons-Months

Time (T) = 2.5 x (31.2055) pow (0.38)

= 2.5 x 3.6966

= 9.2415

T = 9 Months

People Required (R) = E/T

R = 31.2055/9.2415

R = 3.3744

R = 3 persons

Experiment – 1.d: Perform Estimation of effort using FP Estimation for chosen system.

Function Point (FP) estimation is a method used to estimate the size, effort, and time required for software development projects. Here are the steps to perform FP estimation for your find your opportunity system:

Step 1: Identify the Functional User Requirements

1. **External Inputs (EI)**: These are the inputs into the system. For the find your opportunity, these could be:
    - User Registration
    - Task Creation
    - Task Update
    - Task Deletion
    - Project Creation
    - Category Creation
2. **External Outputs (EO)**: These are outputs from the system. For the find your opportunity, these could be:
    - Task List Display
    - Project List Display
    - Notification Display
3. **External Inquiries (EQ)**: These are inquiries where the user receives data but no significant processing occurs. For the find your opportunity, these could be:
    - Task Search
    - Project Search
4. **Internal Logical Files (ILF)**: These are logical groups of data maintained within the system. For find your opportunity, these could be:
    - User Data
    - Task Data
    - Project Data
    - Category Data
    - Notification Data
5. **External Interface Files (EIF)**: These are logical groups of data used by the system but maintained by another application. For the find your opportunity, these might not be applicable if there is no integration with external systems, or they could be:
    - External Calendar Integration

Step 2: Assign Complexity Weights

Assign weights based on the complexity (low, average, high) of each type of function.

Function Type	Weight (Low)	Weight (Average)	Weight (High)
External Inputs (EI)	3	4	6
External Outputs (EO)	4	5	7
External Inquiries (EQ)	3	4	6
Internal Logical Files (ILF)	7	10	15
External Interface Files (EIF)	5	7	10

Step 3: Calculate Unadjusted Function Points (UFP)

Identify the count of each function type and assign complexity.

- **External Inputs (EI):**
  - User Registration (Average) = 1 * 4 = 4
  - Task Creation (Average) = 1 * 4 = 4
  - Task Update (Average) = 1 * 4 = 4
  - Task Deletion (Average) = 1 * 4 = 4
  - Project Creation (Average) = 1 * 4 = 4
  - Category Creation (Average) = 1 * 4 = 4
- **External Outputs (EO)**:
  - Task List Display (Average) = 1 * 5 = 5
  - Project List Display (Average) = 1 * 5 = 5
  - Notification Display (Average) = 1 * 5 = 5
- **External Inquiries (EQ):**
  - Task Search (Average) = 1 * 4 = 4
  - Project Search (Average) = 1 * 4 = 4
- **Internal Logical Files (ILF):**
  - User Data (Average) = 1 * 10 = 10
  - Task Data (Average) = 1 * 10 = 10
  - Project Data (Average) = 1 * 10 = 10
  - Category Data (Average) = 1 * 10 = 10
  - Notification Data (Average) = 1 * 10 = 10
- **External Interface Files (EIF):**
  - External Calendar Integration (Low) = 1 * 5 = 5

Step 4: Compute the Unadjusted Function Points (UFP)

UFP=∑(count× weight)\text{UFP} = \sum (\text{count} \times \text{weight})UFP=∑(count× weight)
UFP=4+4+4+4+4+4+5+5+5+4+4+10+10+10+10+10+5=102\text{UFP} = 4 + 4 + 4 + 4 + 4 + 4 + 5 + 5 + 5 + 4 + 4 + 10 + 10 + 10 + 10 + 10 + 5 =
102UFP=4+4+4+4+4+4+5+5+5+4+4+10+10+10+10+10+5=102

Step 5: Calculate the Value Adjustment Factor (VAF)

Rate the 14 General System Characteristics (GSCs) on a scale of 0 to 5 and compute the VAF.

For simplicity, let's assume an average complexity rating of 3 for all GSCs.

Total Degree of Influence=14×3=42\text {Total Degree of Influence}
= 14 \times 3
= 42 Total Degree of Influence
=14×3=42
VAF=0.65+(0.01×Total Degree of Influence) =0.65+(0.01×42)
=0.65+0.42=1.07\text {VAF} = 0.65 + (0.01 \times \text {Total Degree of Influence})
= 0.65 + (0.01 \times 42) = 0.65 + 0.42
=1.07VAF=0.65+(0.01×Total Degree of Influence)
=0.65+(0.01×42)
=0.65+0.42
=1.07

Step 6: Calculate the Adjusted Function Points (AFP)

AFP=UFP×VAF
=102×1.07=109.14\text {AFP}
= \text {UFP} \times \text {VAF}
= 102 \times 1.07
= 109.14AFP
=UFP×VAF
=102×1.07
=109.14

Step 7: Estimate Effort

To estimate the effort, you need to determine the productivity rate, which is typically measured in function points per person-month. Let's assume an average productivity rate of 20 function points per person-month.

Effort=AFP Productivity Rate

=109.1420≈5.46 person-months\text {Effort}

= \frac {\text {AFP}} {\text {Productivity Rate}}

= \frac {109.14}{20} \ approximately 5.46 \text {person-months}

Effort = Productivity Rate AFP

=20109.14

≈5.46 person-months

Experiment – 1.e: Analyze the Risk related to the project and prepare RMMM plan.

Risk	Potential Impact	Mitigation Strategy	Monitoring Approach	Management Action
Data Security Breaches	Unauthorized access to sensitive user data, compromising privacy and trust.	Implement end-to-end encryption for data transmission and storage. Conduct regular security audits. Ensure compliance with data protection regulations (e.g., GDPR).	Monitor access logs, conduct periodic security audits, and use intrusion detection systems (IDS).	Immediate response plan for breaches, notify affected users, and improve security measures based on audit findings.
Model Accuracy and Reliability	Inaccurate predictions leading to unreliable energy usage recommendations.	Use robust data preprocessing techniques. Validate models with real-time data and user feedback. Implement fallback mechanisms for degraded model performance.	Continuous monitoring of prediction accuracy and performance metrics.	Implement model retraining schedule based on performance metrics and user feedback.
Technological Dependencies	Dependency on external APIs or platforms prone to downtime or performance issues.	Use API monitoring tools to track uptime and performance. Develop backup data retrieval methods and error handling strategies.	Monitor API performance metrics and receive alerts for downtime or performance degradation.	Have contingency plans for switching APIs or alternative data retrieval methods.
User Engagement and Adoption	Low user adoption rates or disengagement from the energy tracking platform.	Conduct user surveys and feedback sessions during development. Design intuitive and user-friendly interfaces. Provide personalized energy-saving tips and incentives.	Analyze user engagement metrics (e.g., active users, session duration, feature usage).	Iterate on UI/UX design based on user feedback. Plan marketing campaigns to promote platform benefits and features.
Scalability Challenges	Inability of the platform to handle increased data volume or user traffic growth.	Design for scalability using cloud-based infrastructure (e.g., AWS, Azure). Implement caching mechanisms and load balancing.	Perform load testing simulations to identify system bottlenecks and scalability limits.	Scale infrastructure resources based on load testing results and anticipated growth projections.

Regulatory Compliance	Failure to comply with energy usage regulations and data protection laws.	Stay updated with regulatory requirements (e.g., energy consumption standards, data privacy laws). Implement features for user consent and data anonymization.	Regularly review and update compliance measures based on regulatory changes.	Conduct audits to ensure adherence to regulatory standards and update policies as needed.

Risk Monitoring and Review:

Regularly review identified risks, report on their status, and develop contingency plans for high-impact risks. Implement a system for continuous risk assessment and communication with stakeholders.

Estimated Cost:

The estimated cost associated with addressing these risks depends on various factors such as the resources required for mitigation strategies, potential impact on project timelines, and the complexity of risk management activities. It's essential to allocate a contingency budget to cover unforeseen expenses related to risk mitigation.

Overall, incorporating risk analysis, RMMM planning, and estimated costs into the project budget ensures that potential risks are proactively managed, minimizing their impact on project success and financial outcomes.

Experiment – 1.f: Develop Time-line chart and project table using PERT or CPM project scheduling methods.

To estimate the total cost for RMMM, we will sum up the costs for each mitigation strategy

Let's assumed cost for RMMM is $90,000

RMMM Cost Estimation = Cost Estimation of each risk/Assumed Cost Of RMMM

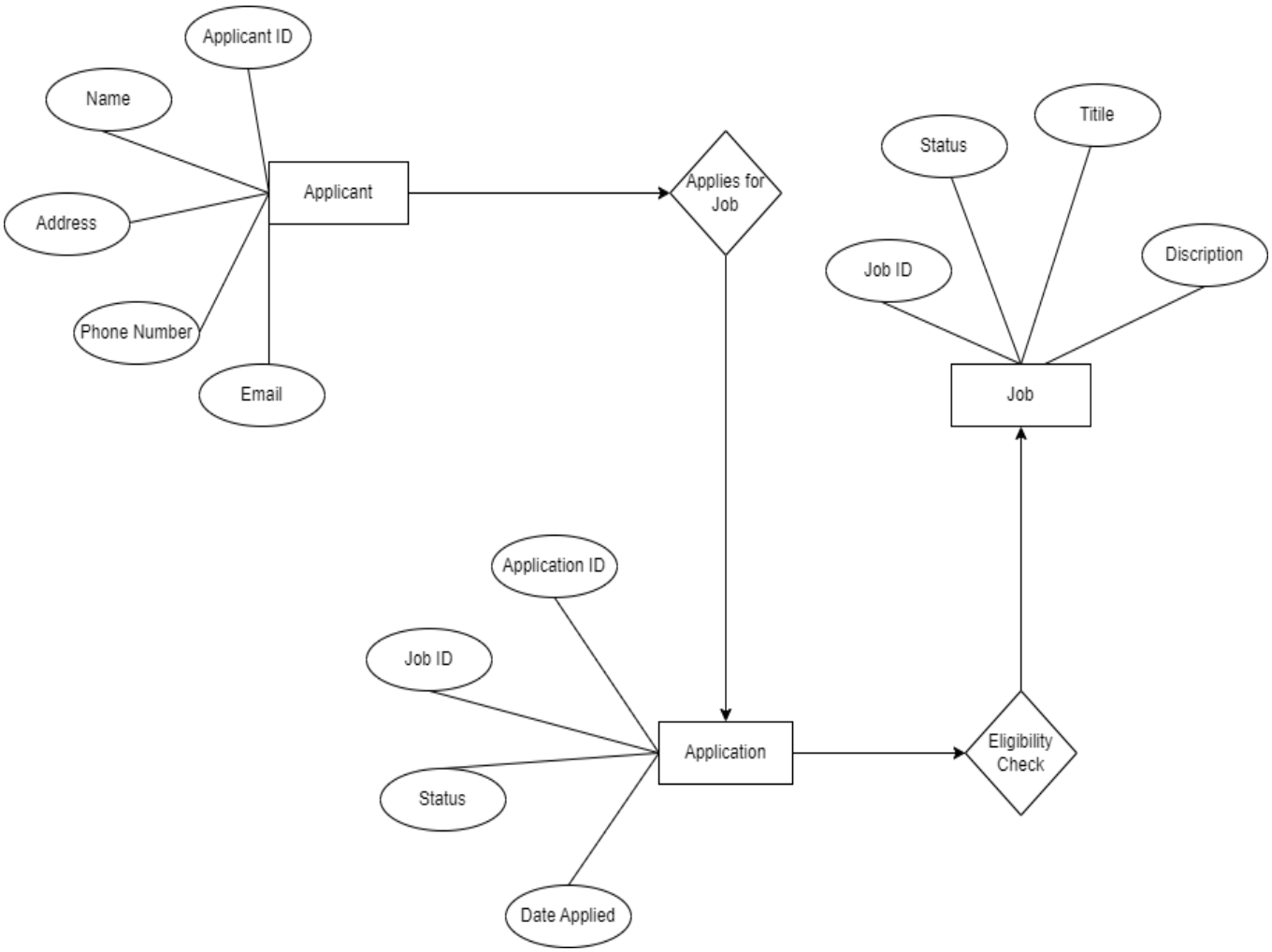
$$\text{RMMM} = \frac{\$50000 + \$10000 + \$8000 + \$7000 + \$12000 + \$6000 + \$9000 + \$4000 + \$8000 + \$3000 + \$10000 + \$7000}{\$90,000}$$

RMMM = $ 89,000 / $ 90,000

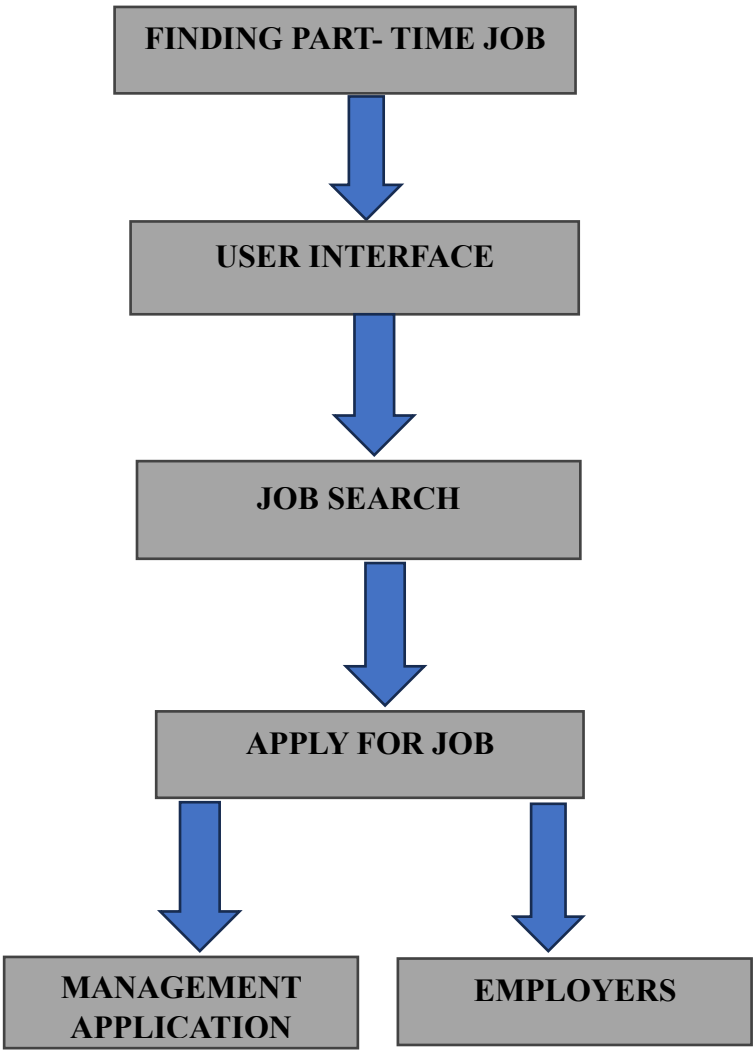
RMMM = 0.9888

Experiment – 1.g: Draw E-R diagrams, DFD, CFD and structured charts for the project.

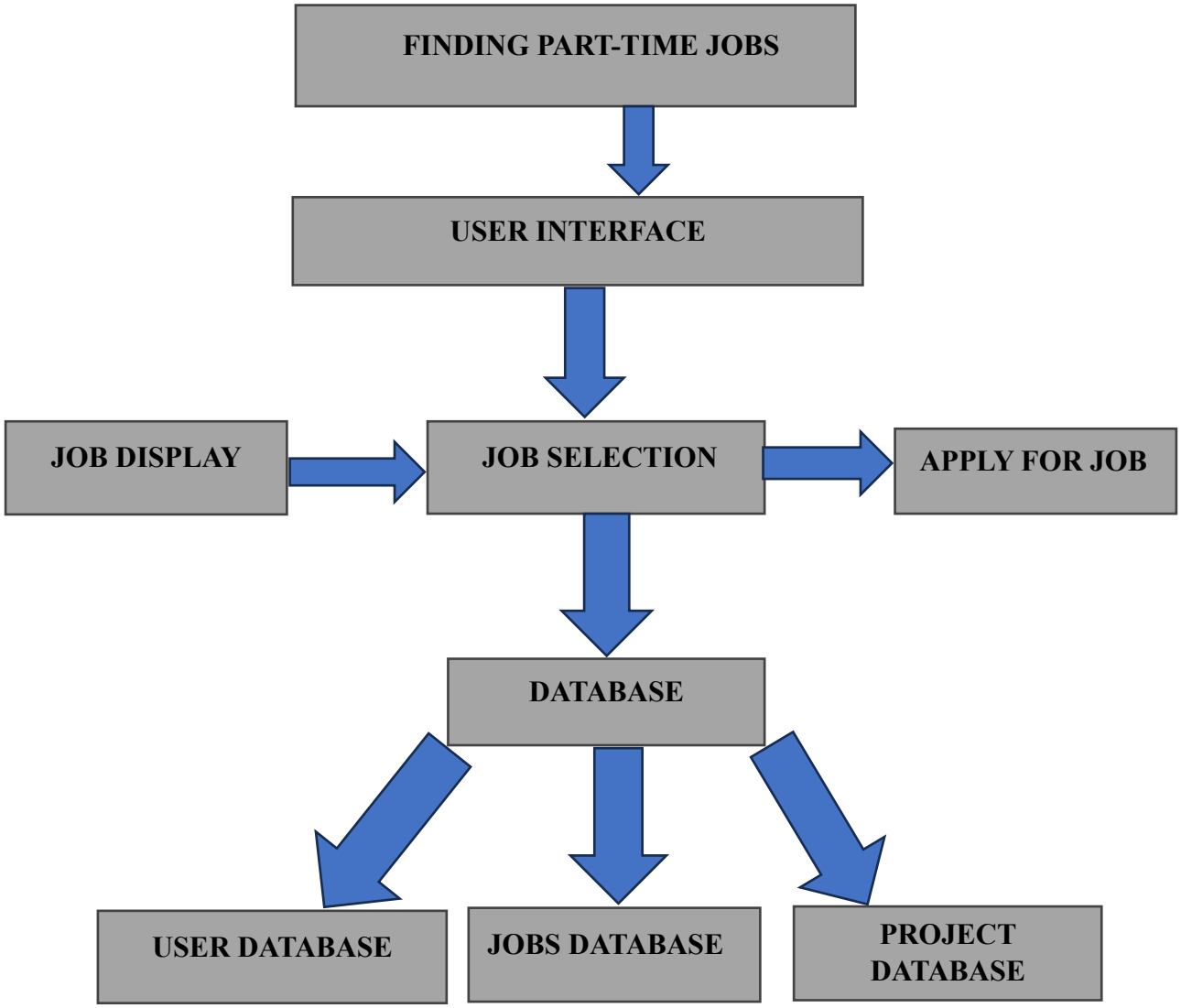
ER DIAGRAM



DFD-Level 0


FINDING PART- TIME JOB
USER INTERFACE
JOB SEARCH
APPLY FOR JOB
MANAGEMENT APPLICATION
EMPLOYERS

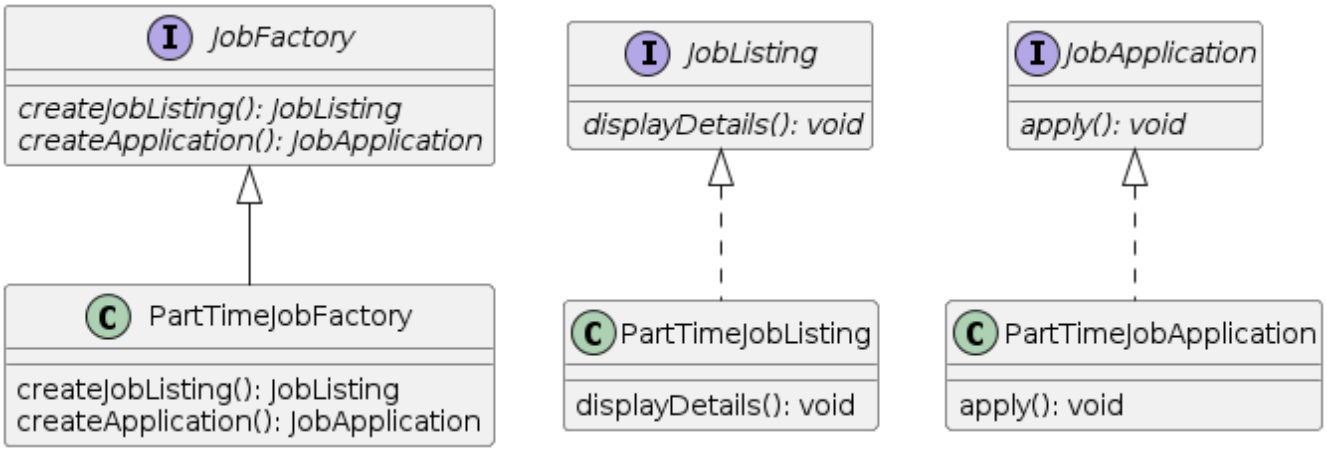
DFR-Level 1


FINDING PART-TIME JOBS
USER INTERFACE
JOB DISPLAY
JOB SELECTION
APPLY FOR JOB
DATABASE
USER DATABASE
JOBS DATABASE
PROJECT DATABASE

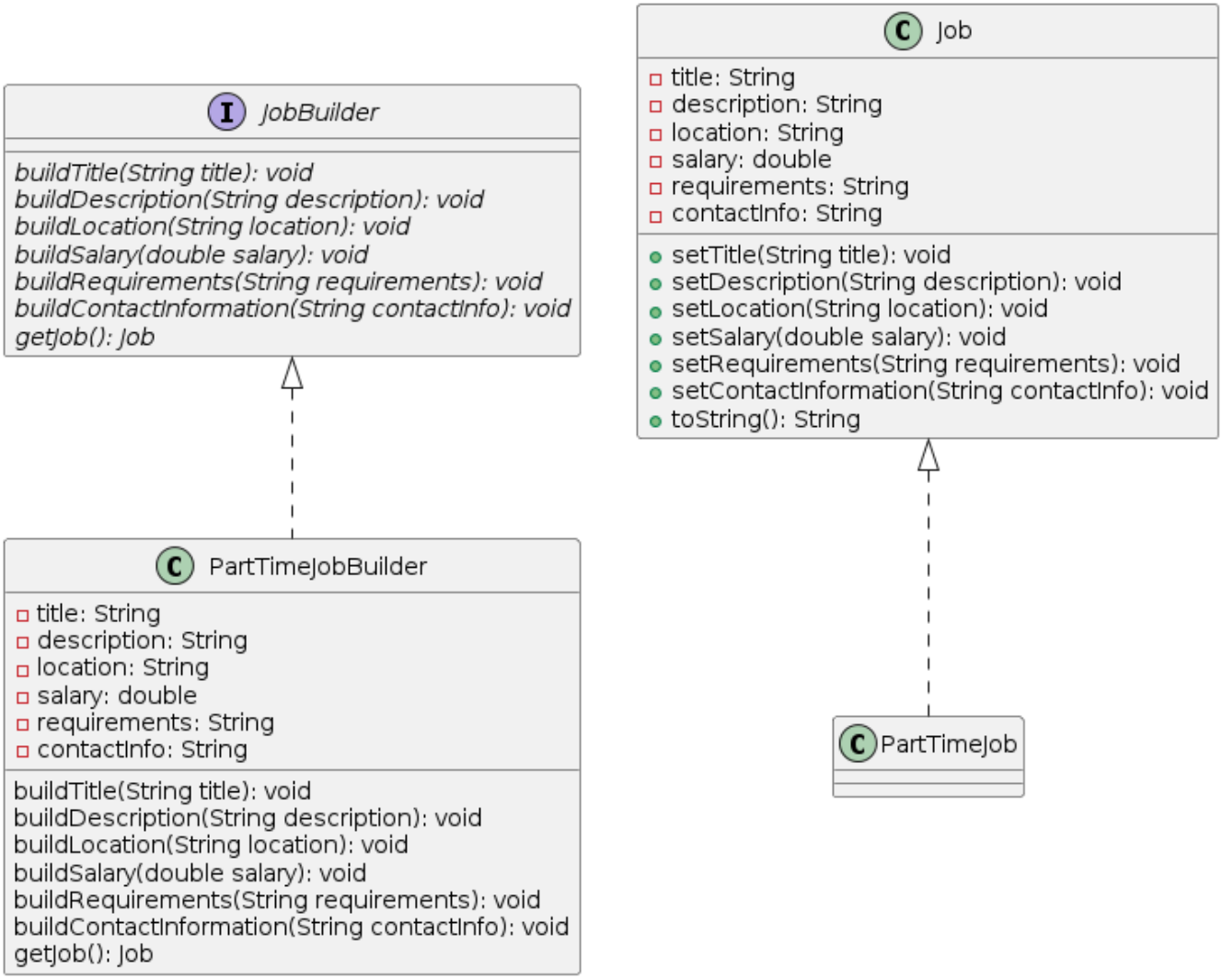
Experiment – 1.h: Design of Test cases based on requirements and design.

Module	Test Case ID	Description	Preconditions	Test Steps	Expected Results	Status
User Management	UM001	Register with valid details	None	1. Open app 2. Go to registration 3. Enter valid details 4. Submit	User account created	Pass
User Management	UM002	Register with invalid details	None	1. Open app 2. Go to registration 3. Enter invalid details 4. Submit	Error message displayed	Pass
Job Management	JM003	Delete a job	Job exists	1. Go to job management 2. Select job 3. Delete	Job deleted	Pass
Job Search	JS001	Search jobs with valid criteria	Jobs are available	1. Go to job search 2. Enter valid criteria 3. Search	Matching jobs displayed	Pass
Job Search	JS002	Search jobs with invalid criteria	None	1. Go to job search 2. Enter invalid criteria 3.	No jobs found message	Pass
User Management	UM004	Login with invalid credentials	User account exists	1. Open app 2. Enter invalid credentials 3. Click login	Error message displayed	Pass
User Management	UM005	Logout	User is logged in	1. Click logout	User logged out	Pass
Job Management	JM001	Post a new job	User is logged in as employer	1. Go to job posting 2. Enter job details 3. Submit	Job posted	Pass
Job Management	JM002	Edit a job	Job exists	1. Go to job management 2. Select job 3. Edit details 4. Save	Job updated	Pass
Management		job	in as job seeker	details 2. Click apply 3. Submit application	submitted	
Application Management	AM002	View applied jobs	User is logged in as job seeker	1. Go to applied jobs	Applied jobs listed	Pass
Logging	LG001	View user activity log	User is logged in	1. Go to activity log	Activity log displayed	Pass
Logging	LG002	View job application log	Admin is logged in	1. Go to application log	Application log displayed	Pass

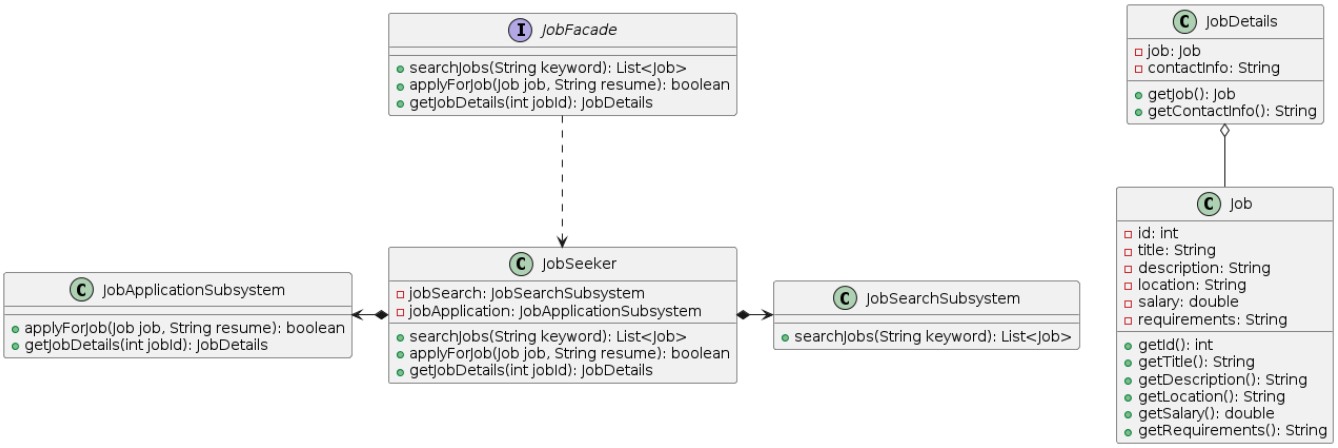
Experiment – 2: Using UML/JAVA, design Abstract Factory design pattern.



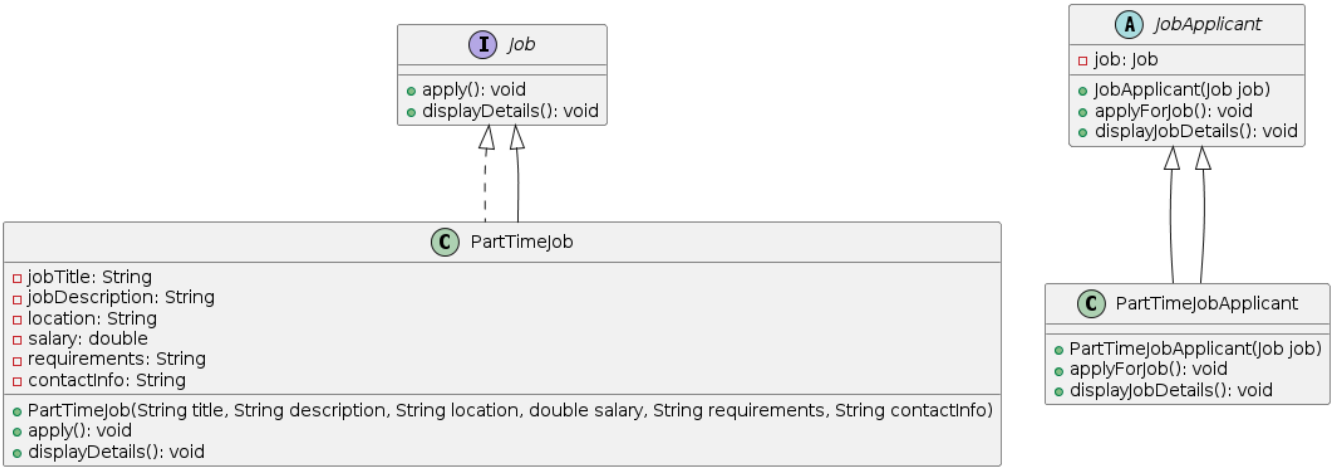
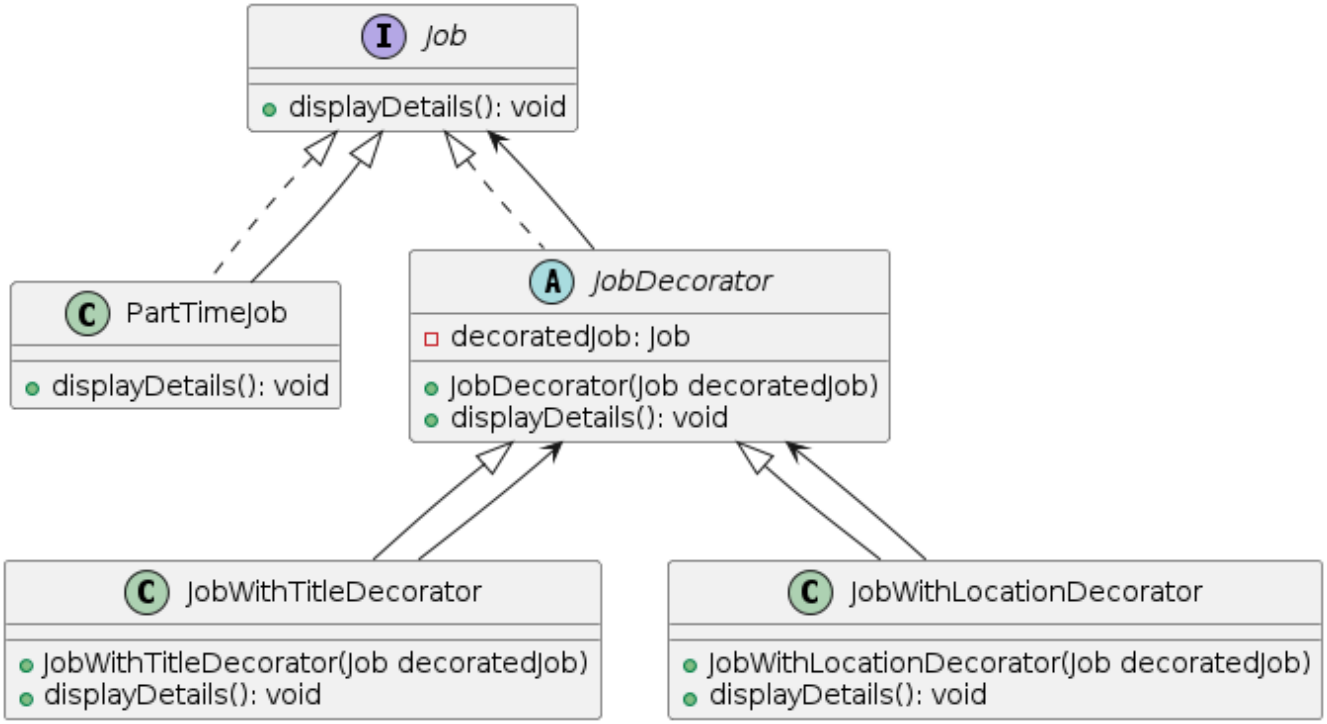
Experiment – 3: Using UML/JAVA, design Builder design pattern.



Experiment – 4: Using UML/JAVA, design facade design pattern.



Experiment – 5: Using UML/JAVA, design Bridge design pattern.



Experiment – 6: Using UML/JAVA, design Decorator design pattern.